Syntactic height impacts prosodic size: An argument for cyclic prosodification*

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1. Introduction

In this paper, we will put forward an argument for the claim that the syntax-prosody mapping must apply cyclically in a bottom-up fashion. Smaller prosodic categories are constructed before bigger prosodic categories and it is the presence of specific cyclic nodes that trigger the construction of a prosodic category of a given size. Our argument is based on two robust crosslinguistic generalizations about different phenomena relating to the morphosyntax-prosody interface. The two phenomena we will be concerned with are full and partial reduplication expressing verbal number and phonologically determined second position placement of clausal coordinators.

Despite the fact that these two phenomena are arguably unrelated and operate on structures of different sizes, we see that the two patterns concerning the morphosyntax-prosody mapping converge on the same abstract generalization given in (1):

(1) Correlation of Syntactic Height and Prosodic Size (COSHAPS):
If two syntactic heads (or their respective exponents) trigger the same prosodic process on an element X, then the process of the syntactically higher head will apply to a bigger (or equal) prosodic domain within X.

Let us illustrate the workings of (1) using the tree in (2) below. In the structure in (2), both heads A and B trigger the same type of phonological process, which makes reference to the prosodic constituency of their complement domain. If that is the case then, the lower head B will only be able to reference the smaller prosodic domain Δ while the higher head A will have access to the bigger prosodic category $\Delta+1$.

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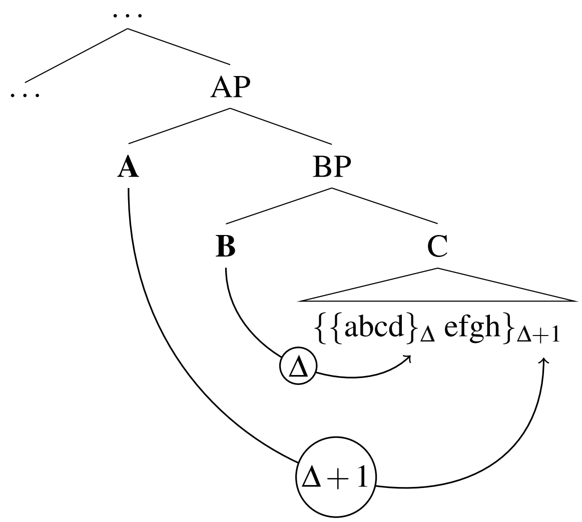
(2) The Correlation of Syntactic Height and Prosodic Size:



As noted above, we will show evidence from two crosslinguistic studies of unrelated phenomena that this generalization shows its effects in two completely unrelated empirical domains of the morphosyntax-prosody interface. Building on that, we go on to argue that the most natural explanation for this generalization is that prosodic constituents are built cyclically in the course of the derivation. If, in the structure in (2), the prosodic domain $\Delta+1$ is only built *after* the completion of BP, it falls out naturally that B will not have access to it. In that case, we can simply maintain the generalization that the prosodic processes triggered by B and A always refer to the biggest prosodic constituent that is accessible at a given point in the derivation.

The paper is structured as follows: Section 2 will present the argument from verbal number reduplication and Section 3 will present the argument from second position placement of clausal coordinators. After an interim summary in Section 4, Section 5 will will provide a a prosodic-cyclic analysis of verbal number reduplication and sketch an account of second position coordinators before concluding in Section 6.

2. Flavors of verbal number

2.1 Two types of verbal number

While nominal number is concerned with the number of entities, verbal number describes the number of events involved in an utterance. This can be observed in two flavours, called *event number* and *argument number* (Corbett 2000).[1] Argument number can be observed in the Kipsigis example in (3).

[1] Usually, the term *participant number* is used instead of *argument number*, see for example Corbett (2000) and Thornton (2020). We changed the terminology to better contrast with the term partial reduplication and both terms can be understood as being interchangable.

(3) Kipsigis argument number (Kouneli 2020:7)

a. lɑbɑt-i làːkwɛ̀ːt.
run.**SG**-IPFV child.NOM
'The child is running.' (baseline)

b. ruaj lɑ̀ːgôːk.
run.**PL**.IPFV children.NOM
'The children are running.' (argument number)

In (3a) and (3b), different verb forms are used to express meaning 'to run'. This difference is due to the number of arguments involved in the sentences. This suppletion pattern is related to a phenomenon called *argument number*: a multiplicity of events is distributed across several arguments. Argument number usually only applies to a subset of verbs in a language and is often observed in addition to other forms of number agreement (Corbett 2000). In Kipsigis, for example, only three verbs supplete for number, as seen in (3). Additionally, suffixes mark regular number agreement independently (Toweett 1979). See (4), where one of the forms of 'to run' appears with first person singular agreement.

(4) keː-**ɑ**-lɑbɑt-ɑt-e
to-1SG-run.SG-AMB-IPFV
'I was running around.'

In event number, as in (5), on the other hand, the verb form *bet* 'to split' is reduplicated. This is not due to the number of arguments but due to the expressed meaning 'again and again': A multiplicity of events that may involve one or several arguments. Event number is known to be productive and available to all verbs in a given language (Corbett 2000).

(5) Kipsigis event number

a. kiː-ɑ-bet-e kwen-deːt
PST-1SG-split-IPFV wood-**SG**
'I was splitting a piece of wood.'

b. kiː-ɑ-bet∼ɑː∼bet-i kwen-ik
PST-1SG-split∼LK∼split-IPFV wood-**PL**
'I was splitting many pieces of wood **again and again**.'

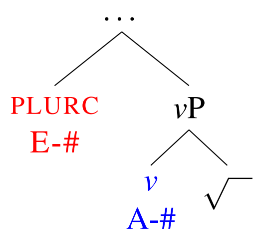
2.2 The correlation

The flavours of verbal number are argued to be located in different syntactic positions. Argument number is very close to the verbal root. It is argued to be located on *v* (Amato 2018, Fiebig in preparation) or lower (Thornton 2020). From this position, the argument number head agrees with a DP (usually the object) to find the relevant number features. Event number, on the other hand, is located at least outside of *v*P, called PLURC here.[2] See

[2] PLURC stands for *Pluractionality*, a different name used for event number, see e.g. Wood (2007).

the structure in (6) for a visual representation of the locations of event number (E-#) and argument number (A-#).

(6)



It is cross-linguistically common for languages to employ reduplication as a marking strategy for verbal number (Mattiola 2019). In the case of Niuean for example, different reduplication types express event and argument number. In the example in (7a), partial reduplication of the verbal root *hala* expresses argument number. The event ('cutting') takes place several times, distributed across several arguments ('branches'). In contrast to that, in (7b), a total reduplication of the verb *noko* expresses event number. The event ('knocking') takes place several times with the same arguments in each instance.[3]

(7) Niuean (Haji-Abdolhosseini et al. 2002)

a. kua ha-hala e ia e tau lā akau.
PERF RED.ARG-cut ERG he ABS PL branch tree
'He cut the branches.' (A#)

b. ne noko-noko e ia e gutuhala
PST RED.EVENT-knock ABS she ABS door
'She knocked on the door (many times).' (E#)

This correlation of reduplication size and type of verbal number does not seem to be limited to Niuean. A typological investigation reveals a cross-linguistic correlation: Partial reduplication marks both flavours of verbal number, but full reduplication marks only event number. In the following, we will report the results of a typological investigation into reduplication as a marker for verbal number cross-linguistically.

2.3 The typology

Fiebig (in preparation) investigates 67 languages from 37 families with respect to reduplication as a marker for verbal number. This study reveals 6 possible types of languages concerning reduplication as a spell out of event and argument number. These types can be found in Table 1.

The table shows that there are six different ways of how reduplication can express the types of verbal number. Some languages employ only one type of reduplication to express one type of verbal number. Class 2, for example, employs partial reduplication to express event

[3] Haji-Abdolhosseini et al. (2002) actually observe several more types of reduplication with specific meanings expressed by them.

class	partial reduplication	full reduplication	# of L
1	-	event-#	23
2	event-#	-	11
3	argument-#	-	2
4	argument-#	event-#	6
5	argument-# and event-#	-	23
6	event-#	event-#	2
	any	argument-#	unattested

Table 1: Typology of reduplication types spelling out verbal number

number. The languages in class 3, on the other hand, employ partial reduplication to express argument number. There are also languages like that in class 4, where partial reduplication expresses argument number while full reduplication expresses event number. Niuean, for example, belongs to this class, as we already saw in (7). This table also shows that there is one potential type of languages which is unattested: there is no language in the sample that marks argument number by full reduplication. Table 2 visualizes this observation.

	partial reduplication	full reduplication
Event number	✓	✓
Argument number	✓	✗

Table 2: Which types of reduplication are able to express which type of verbal number

As the first line indicates, event number can be expressed by either partial or full reduplication. The second line reveals the gap: argument number is only expressed by partial and not by full reduplication cross-linguistically. This observation is only limited to reduplication as a spell-out of verbal number though. This makes no statement about other marking strategies for verbal number or other grammatical categories expressed by reduplication. This observation can be summarized as in (8).

(8) Generalization:
Partial reduplication expresses event- and argument number, but full reduplication only ever marks event number. Full reduplication never marks argument number.

3. Shifting coordinators

3.1 Two types of coordinators

Many languages have some morphosyntactic elements used to coordinate two constituents. These elements, also known as coordinators, come in two types (see Haspelmath 2007). Some languages like English use one coordinator acting as a linker in between the two constituents they coordinate. This is known as the monosyndetic pattern. Other languages use one coordinator per conjunct, which is known as the polysyndetic pattern. The difference is illustrated with noun phrase coordination from the Caucasian language Avar:

(9)
a. [keto gi] [ħve gi]
cat ADD dog ADD

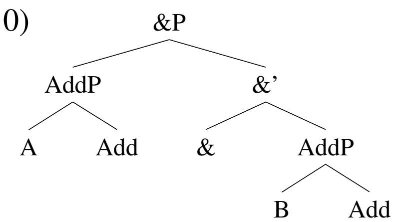
b. [keto] va [ħve]
cat and dog
'cat and dog'

c. [keto gi] va [ħve gi]
cat ADD and dog ADD
'cat and dog'
Mitrović and Sauerland (2014)
(bracketing ours)

Speakers of Avar can either attach a so-called additive marker *gi* to each conjunct, they can use the linker *va* in between the conjuncts or they can combine the strategies. In terms of structure, it has been argued that the linker *va* is the realization of the coordination head $\&^0$ while the polysyndetic coordinators *gi* are additive focus markers that attach to each conjunct before they are linked (see Bhatt 2014, Mitrović and Sauerland 2014).

We can thus assume a coordination structure with two conjuncts A and B as in 10:

(10)



3.2 The correlation

Weisser (2024) investigates non-canonical placement patterns of clausal coordinators and finds quite a number of cases where the coordinators do not appear in the position where we would expect them to be given the structure in (10). Instead they appear, in the vast majority of cases, in some sort of a second position inside their syntactic complement. In some cases this position is defined in syntactic terms (i.e. the coordinator appears after the first XP of the conjunct) but in some cases, it is also defined in prosodic terms. Then, the coordinator appears after either the first phonological word (ω) or the first phonological phrase (ϕ). An example of the former is Latin (11), an example of the latter is Yorùbá (12).

(11)
a. ... sine=**que** ferr-o fu-erint
without=COORD iron-ABL be-SBJ.PERF.3PL
'... and that they were without swords' Latin, Embick (2007:309)

b. ... **que** $\{\text{sine}\}_\omega$ $\{\text{ferro}\}_\omega$ $\{\text{fuerint}\}_\omega$

(12)
a. ... Ọlá ò bá **sì** lọ.
... Ola should have COORD go
'...and Ola should have gone.' Yorùbá, Aremu and Weisser (2024)

b. ... **sì** $\{\text{Ọlá ò bá}\}_\phi$ $\{\text{lọ}\}_\phi$

The connection to the abstract pattern discussed in the introduction is that in the sample in Weisser (2024), there is a clear correlation between the syntactic type of the coordinator (monosyndetic vs polysyndetic) and the shifting pattern. When syntactic polysyndetic coordinators shift across a prosodic constituent, they move to the first phonological word (ω). Monosyndetic coordinators shift after the first phonological phrase (ϕ).

(13)

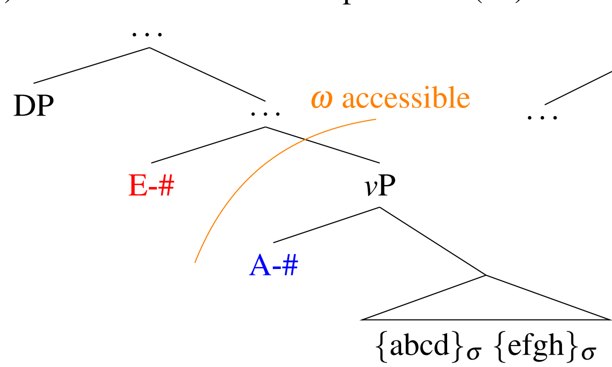
	[ω-COORD	[ϕ-COORD
ADD	Latin, Cherokee, Kalaallisut, Ancient Greek Hittite, Gaulish, Gothic, Old Irish, Coptic	–
&	–	Yorùbá, German, Polish

Under the assumption that phonologically determined second position elements originate in initial position,[4] we can rephrase the correlation above in familiar terms: The lower syntactic head (ADD) skips a smaller prosodic constituent (ω) whereas the higher syntactic head ($\&^0$) skips a bigger prosodic constituent (ϕ). Below, we will now compare the two abstract configurations and then work our way towards an explanation.

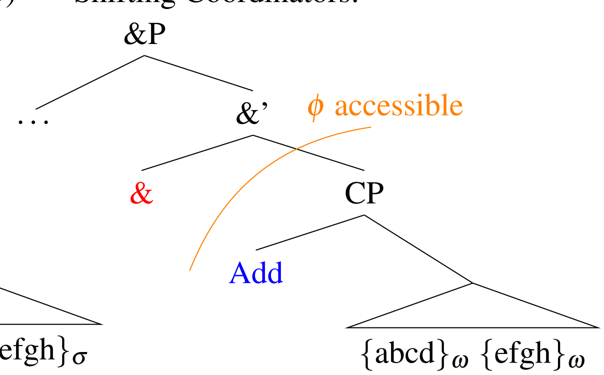
4. Interim Summary

In the previous two sections, we saw two case studies of processes referring to phonological domains of different sizes. The striking result in both case studies was that we identified two syntactic heads triggering the same process. Even though these two heads are, syntactically speaking, quite local to each other, the processes they trigger robustly affect phonological domains of different sizes. Crucially, in both case studies, the differences both pattern the same: The process triggered by the lower syntactic head refers to the smaller phonological domain whereas the higher syntactic head refer to the bigger phonological domain. The trees below emphasize the abstract similarities:

(14) Verbal Number Redupl:



(15) Shifting Coordinators:



The intuition we will follow below is that it is not a coincidence that, in both cases, the two heads in question are separated by the nodes CP and vP, which are the standardly

[4]See e.g. Marantz (1988), Halpern (1995), Embick and Noyer (2001), Weisser (submitted) for arguments to this conclusion.

assumed cyclic nodes in phase-based Minimalist syntax. We will assume a cyclic model of syntax-prosody mapping where, upon completion of a cyclic node, the prosodic constituents in that domain are generated. In other words, only at the point in the derivation where a cyclic node like vP or CP is reached, the prosodic constituents in that domain will become accessible.

5. Deriving the patterns

5.1 Assumptions

For the analysis, we follow Kratzer and Selkirk (2007), Pak (2008), Newell (2008), Kahnemuyipour (2009), and Fenger (2020) in assuming that syntactic domains delimit which prosodic structure is built. In addition, we assume that prosodic structure is built incrementally, as proposed by Inkelas (1990) and Bermudez-Otero (2019). Vocabulary insertion may occur bottom-up (Bobaljik 2000) and when vocabulary insertion of a node takes place, the exponent is linearized with respect to previously generated structure (Kalin 2022, Weisser 2024). We further assume that operations such as Reduplication or the second position dislocation operation INTEGRATE are sensitive to the prosodic structure built at that point.

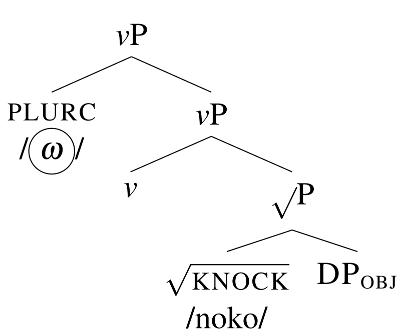
5.2 Example derivation

As observed earlier on, event number is expressed by full and partial reduplication. Argument number, on the other hand, is only expressed by partial reduplication and cannot be marked by full reduplication. The following derivations explain this typological gap based on the assumptions given above.

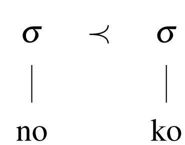
5.2.1 Full reduplication

In the case of total reduplication (event number), the relevant head is an adverbial element PLURC, which is located above *v*P. The head is spelled out by a floating prosodic word (ω) that is eventually responsible for the total reduplication. Consider the tree in (16) that represents the (simplified) structure for event number in Niuean, as seen in (7b).

(16)

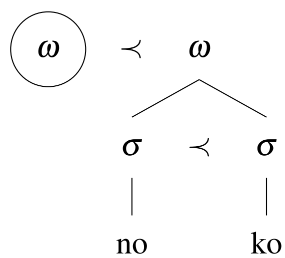


(17)



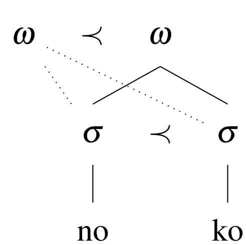
In a cyclic, bottom-up process of vocabulary insertion, the first cycle of spell-out takes place.[5] The first phase head, *v*, triggers the spell-out of itself and its complement (see Marantz 2001 and Creemers et al. 2018 on *v* being a phase head and Holmberg 1999 and Fenger 2020, and references therein, on the head being included in the spell-out domain it triggers). Note though, that little *v* is not being spelled out overtly in this case. The inserted vocabulary items of this phase is linearized. After spell-out and linearization, an optimization step takes place. This optimization process is less relevant in this case but will become more important later on. It will be this optimization step (just taking place in another cycle) that leads to reduplication as a form of repair. The result of these steps can be seen in (17). After this cycle of spell-out the next cycle takes place. Within this next cycle, the heads in the existing structure is mapped onto prosodic words. Additionally, in this cycle, vocabulary insertion for PLURC takes place and the floating prosodic word is linearized with respect to the result of the first cycle. The structure in (18) shows this point in the derivation.

(18)

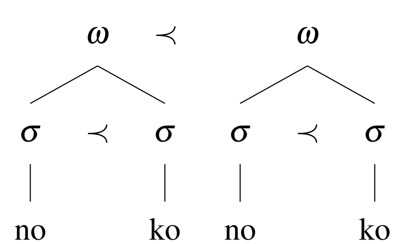


Now the optimization step becomes relevant since it takes care of the floating element. Ideally, the floating prosodic word should associate to the content of the already existing syllables (as sketched in 19), but this is not optimal due to the crossing association lines this would create. In this scenario, reduplication is the cheapest repair. It violates no DEP or MAX constraints, as epenthesis of some material or deletion of the floating element would (see Kirchner 2010 on the details of a prosodic approach to reduplication.). Reduplication still allows to associate the floating element to prosodic material.

(19)



(20)



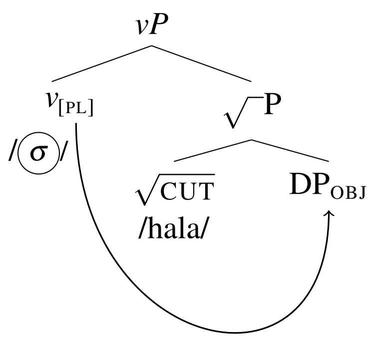
The result of this full cycle of spell-out (including optimization) in (20) is now fed to the next cycle of spell out, where the next vocabulary items are inserted. At this point, the whole *v*P will be mapped onto prosodic structure to create a prosodic phrase.

[5] We are glossing over the actual first cycle here and in the following structures. A cycle, spelling out the object DP precedes all cycles shown here.

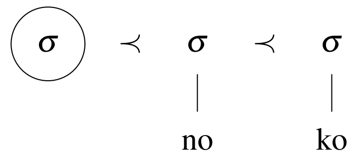
5.2.2 Partial reduplication

For partial reduplication in argument number, an agreement operation between little v, the categorizing head (probe) and the object DP (goal) precedes any spell out. In this agreement step, the number features of the object DP are transferred to the *v* head. This is shown in (21), which illustrates the structure for argument number as in the Niuean example in (7a).

(21)

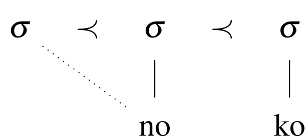


(22)

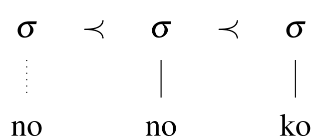


Little *v* now induces the first cycle of spell-out. Vocabulary insertion and linearization take place. This time, little *v* is overtly spelled out by a floating syllable. The result is the structure in (22). Now, a step of optimization takes place. Just as in the case of total reduplication, the floating element cannot associate with the existing material; this time because this would create an ambisyllabic segment (see 23). Again, reduplication is the most optimal repair as it violates no other constraints like any epenthesis to fill the syllable or deletion of the floating element would.

(23)



(24)

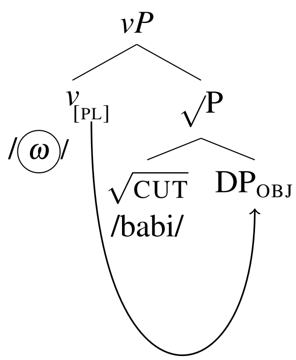


This optimized structure can now be fed to the next cycle where — upon insertion and linearization of the next vocabulary item — the existing structure will be mapped onto a prosodic word.

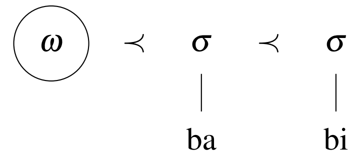
5.2.3 Full reduplication and argument number

Given the analysis of reduplication as a repair strategy, the typological gap observed in section 2.3 can easily be explained. Let us revisit the steps we just saw for argument number in (21-24) but this time assume a prosodic word spells out the result of the same agreement-step. This would be required to derive full reduplication, just as in (18). The potential structure is illustrated in (25)with a hypothetical root *babi*.

(25)



(26)



Again, the first cycle of spell-out is induced by little *v* and triggers vocabulary insertion and linearization for *v* and its complement. The result of this step can be seen in 26. Now, just as in the other cases, an optimization step applies to this structure. In contrast to the other cases, association with the existing phonological material is now in fact the most optimal way to deal with the floating prosodic word since this creates no crossing association lines. Reduplication would actually be less optimal than simple association, since it violates a low-ranked faithfulness constraint INTEGRITY, which assigns violations for segments in the input with more than one correspondent in the input (McCarthy and Prince 1995). See the tableau in (27) for a visualization of this optimization process.[6]

(27)

I: ω σ σ ba bi	*FLOAT	NOCROSSING	DEP	MAX FLOAT	INTEGRITY
☞ a. ω σ σ ba bi					
b. ω σ σ ba bi ~ σ σ ba bi					*!

The constraint INTEGRITY is also violated in all the reduplication processes we saw earlier but — since it is ranked so low — these candidates still win. If a floating prosodic word is introduced in the *v*P cyle, association is the cheapest option since it does not violate any relevant constraint. The output of this optimization process is now fed to the next cycle

[6]The tableau omits other potential candidates. These could include epenthesis that violates DEP or a candidate where the floating element is simple deleted, which violates MAX FLOAT.

of spell-out. We therefore never see argument number being spelled out by full reduplication: The constellation necessary for such a reduplication produces no reduplication since it needs no repair of this form. This difference is crucially derived by the fact that the low head little v is spelled out (and its spell-out is optimized) before the structure is mapped onto a prosodic word. In other words, the height of the relevant head decides over the size of the prosodic structure manipulated by it.

5.3 A note on coordination

In the previous section, we have seen that it was the relative placement of the two heads inside or outside of vP that decided whether full ω-reduplication was a viable strategy. In the case of the clausal coordinators, the crucial node is the CP-node. As can be seen in (15), monosyndetic coordinators (&) sit above CP and polysyndetic ones (ADD) right below it. The second position dislocation rule we adopt here (see Weisser 2024 for a much more elaborate proposal) informally states that a prosodically determined second position element will skip the first constituent inside of the one that it immediately precedes. As CPs are typically associated with the prosodic constituent of an intonation phrase (ι), & precedes ($\prec$) ι in (28) and the first element within ι is ϕ_1. In (29), ADD does not precede a ι as ιs are only built upon completion of the CP and ADD is part of the CP. Add only precedes ϕ and within ϕ, the first element is ω_1. So, that is what ADD attaches to.

(28) $\& \prec [_{\iota}\ \phi_1 \prec \phi_2 \ldots]$

(29) $\text{ADD} \prec [_{\phi}\ \omega_1 \prec \omega_2 \prec \omega_3 \ldots]$

6. Conclusion

In this paper, we have discussed the results of two unrelated typological studies which suggest robust correlations between the specific morphosyntactic properties of a given element and the size of the prosodic domain it affects. The first study showed that the morphosyntactic type of verbal number correlates robustly with the size of the reduplicant. Event number can be expressed with either total or partial reduplication but argument number can only be expressed with the latter. The second study showed that the type of the coordinator (monosyndetic vs polysyndetic) correlates with the size of the prosodic constituent it skips in cases the coordinator is specified as a second position element. Monosyndetic coordinators appear after the first prosodic phrase whereas monosyndetic coordinators appear after the first prosodic word. We argued that these crosslingustically robust correlations are best explained in a timing-based approach where certain prosodic constituents become accessible only in the course of the derivation. We illustrated how this logic works for the first case study where it was precisely the cyclic vP-node that triggered the formation of prosodic constituents inside of it. Only elements above the vP-node thus have access to the prosodic words inside the vP.

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